

1. User Registration and Login

ID	Test case	Expected	Result	Notes
TC-1.1	Register with valid email, display name, and password ≥ 8 chars with uppercase + number	Account created; user redirected to dashboard	Pass	AuthRepository saves token via SharedPreferences on success. Token observed in HomeFragment.
TC-1.2	Register with a duplicate email address	System rejects registration with an error message	Pass	Handled server-side in PHP; client receives error response and displays toast.
TC-1.3	Login with correct credentials	Session token issued; dashboard loads with user plants	Pass	Token passed via Intent extras to HomeFragment; saved to SharedPreferences via saveToken().
TC-1.4	Login with incorrect credentials	Generic error shown without specifying which field is wrong	Warning	Error handling exists but response message content is controlled server-side. Client-side validation not visible in provided code — unclear if the message reveals which field failed.
TC-1.5	Passwords stored as one-way hash; plaintext never persisted	Database contains bcrypt hash only (REQ-1.4)	Fail	Cannot confirm from Kotlin code alone — requires PHP backend review. No client-side evidence of hashing since hashing happens server-side. Documented gap; treat as unverified until PHP is reviewed.

2.Plant Library Search

ID	Test case	Expected	Result	Notes
TC-2.1	Load plant library on PlantAddingActivity open	Default list of plants populates RecyclerView within 5 seconds	Pass	loadPlants() calls Trefle via APICalls.php on onCreate(). RecyclerView adapter updated with results.
TC-2.2	Search for a plant by common or scientific name	Filtered results appear; list updates with each keystroke	Pass	onQueryTextChanged() fires searchPlants() on each character. Results replace the adapter.
TC-2.3	Select a plant from search results	PlantDataActivity opens with plant species passed; full detail page loads	Pass	clickedPlants() fires Intent with plant_species extra. PlantDataActivity calls Perennial via findPlantID() then LoadPlantData().
TC-2.4	Search for a plant not in Trefle or Perennial database	Graceful empty state	Fail	No empty-state handling found in PlantAddingActivity. If API returns zero results, RecyclerView shows nothing with no user feedback. The SRS itself acknowledges this gap (Section 2.5).

3. Personal Plant Collection

ID	Test case	Expected	Result	Notes
TC-3.1	Add a plant to personal collection	Plant POSTed to plant_add.php; "Plant Added!" toast shown; activity closes	Pass	addPlantToDatabase() in PlantDataActivity builds JSON payload with token, name, species, watering, sunlight, image (Base64), and description. POST to plant_add.php confirmed.
TC-3.2	View personal plant collection as scrollable list	All user plants load in PlantFragment RecyclerView	Pass	PlantRepository.loadAllUserPlants() fetches getPlants.php with user token; LiveData observed in PlantFragment.
TC-3.3	Remove a plant from personal collection	Plant deleted server-side; UI updates; watering logs cleared (REQ-3.4)	Pass	deletePlant() in UserPlantActivity calls deletePlant.php with plant_id and token. Activity finishes on success.
TC-3.4	Collection refreshes after returning from UserPlantActivity	PlantFragment reflects deletion without manual refresh	Pass	onResume() in PlantFragment calls loadUserPlants() — ensures list stays current after delete.

TC-3.5	Add plant where Perennial returns no watering benchmark data	Graceful fallback; plant still added or user informed	Warning	LoadPlantData() calls optString("watering_general_benchmark") — if missing or malformed JSON, the JSONObject parse will throw. Try/catch exists but sets liveData to emptyList() with no user-facing message. Plant add button may show "Error loading image" instead of a meaningful error.
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4. Plant Tracking and Watering

ID	Test case	Expected	Result	Notes
TC-4.1	Home dashboard shows only plants due for water today	Plants where daysUntilWater == 0 appear; others excluded	Pass	nextWaterDay() in HomeRepository calculates daysUntilWater using ChronoUnit.DAYS. Only adds plant to result list when daysUntilWater == 0 AND startDate != todaysDate.
TC-4.2	Tap a plant on the dashboard to log a watering event	waterPlants.php called; plant removed from dashboard; "Plant Watered!" toast shown	Pass	clickedPlants() → homeViewModel.plantWatered() → HomeRepository.plantWatered() → removePlant() updates LiveData and posts toast message.
TC-4.3	UserPlantActivity displays days until next watering	Countdown shown in showerTxt field	Pass	nextWaterDay() in UserPlantActivity calculates and sets plantWatering.text. Logic mirrors HomeRepository correctly.

TC-4.4	Plant watered on the same day it was added shows correct countdown	Should not show as needing water immediately	Fail	HomeRepository.nextWaterDay() checks startDate != todaysDate as a guard, which correctly prevents same-day false positives — but UserPlantActivity.nextWaterDay() uses a different branch (returns todaysDate.plusDays(0)) that sets plantWatering.text to the raw interval string rather than a calculated countdown. Inconsistency between the two implementations.
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5. Weather-Based Watering

ID	Test case	Expected	Result	Notes
TC-5.1	App requests location permission on HomeFragment load	System permission dialog shown; on grant, location fetched	Pass	permissionLauncher registered in HomeFragment; on grant calls homeViewModel.fetchLocation().
TC-5.2	Weather icon updates based on current conditions	Correct icon drawn for rain, clouds, clear, mist	Pass	HomeViewModel maps weatherType string to drawable resource (ic_rain, ic_cloud, ic_light). WeatherString LiveData observed in HomeFragment.
TC-5.3	Rainy weather auto-water all plants in collection	rainyDayWatering.php called; dashboard cleared	Warning	rainyWeatherWatering() called when weatherType == "rain"; clears _plants LiveData to emptyList(). However, this triggers every time the app loads and weather is rain — including if it already ran today. No deduplication check exists in the client code.

TC-5.4	Location data not stored persistently (REQ-6.4)	No lat/Ing written to SharedPreferences or database	Fail	HomeViewModel stores location in _location LiveData (in-memory only) — good. However, lat/Ing are passed directly into the URL string of getWeather() and rainyDayWatering() HTTP calls. The URL is logged via Log.e("myapp", url) which writes coordinates to the device logcat. In a production context this is a privacy concern (REQ-6.4).
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6. Security & Session Management

ID	Test case	Expected	Result	Notes
TC-6.1	Session token stored client-side correctly	Token persisted in SharedPreferences; retrieved on each API call	Pass	AuthRepository uses MODE_PRIVATE SharedPreferences ("auth_prefs") to save and retrieve "usertoken". Used by both HomeViewModel and PlantViewModel via authRepository.getToken().
TC-6.2	API keys not exposed in Android client code	No API keys present in Kotlin source files	Pass	All external API calls (OpenWeather, Perennial, Trefle) routed through APICalls.php on the PHP backend. No API key strings found in provided Kotlin files.
TC-6.3	Session expires when app is closed (REQ-5.3 §5)	Re-authentication required on next app launch	Fail	Token is saved to SharedPreferences which persists across app restarts. The token is never cleared on app close or in any lifecycle method in the provided code. The SRS requires session expiry on close — this is not implemented client-side.

TC-6.4

All
network
calls use
HTTPS

No HTTP URLs in
codebase

Warning

All URLs in the provided files use
http://10.0.2.2:8888 — this is the Android emulator
localhost address used during development.
Acceptable for development/testing only. Must be
replaced with an HTTPS production URL before
deployment. The SRS requires TLS 1.2+.